

Solutions to Statistical Physics of Fields
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Contents

1	Collective behavior
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2

Chapter 1

Collective behavior, from particles to fields

Problem 1.1

The binary alloy: A binary alloy (as in β -brass) consists of N_A atoms of type A, and N_B atoms of type B. The atoms form a simple cubic lattice, each interacting only with its six nearest neighbors. Assume an attractive energy of $-J$ ($J > 0$) between like neighbors A–A and B–B, but a repulsive energy of $+J$ for an A–B pair.

- (a) *What is the minimum energy configuration, or the state of the system at zero temperature?*
- (b) *Estimate the total interaction energy assuming that the atoms are randomly distributed among the N sites; i.e. each site is occupied independently with probabilities $p_A = N_A/N$ and $p_B = N_B/N$.*
- (c) *Estimate the mixing entropy of the alloy with the same approximation. Assume $N_A, N_B \gg 1$.*
- (d) *Using the above, obtain a free energy function $F(x)$, where $x = (N_A - N_B)/N$. Expand $F(x)$ to the fourth order in x , and show that the requirement of convexity of F breaks down below a critical temperature T_c . For the remainder of this problem use the expansion obtained in (d) in place of the full function $F(x)$.*
- (e) *Sketch $F(x)$ for $T > T_c$, $T = T_c$, and $T < T_c$. For $T < T_c$ there is a range of compositions $|x| < x_{sp}(T)$ where $F(x)$ is not convex and hence the composition is locally unstable. Find $x_{sp}(T)$.*
- (f) *The alloy globally minimizes its free energy by separating into A-rich and B-rich phases of compositions $\pm x_{eq}(T)$, where $x_{eq}(T)$ minimizes the function $F(x)$. Find $x_{eq}(T)$.*
- (g) *In the (T, x) plane sketch the phase separation boundary $\pm x_{eq}(T)$; and the so-called spinodal line $\pm x_{sp}(T)$. (The spinodal line indicates onset of metastability and hysteresis effects.)*

Solution:

(a)

Problem 1.2

The Ising model of magnetism: The local environment of an electron in a crystal sometimes forces its spin to stay parallel or anti-parallel to a given lattice direction. As a model of magnetism in such materials we denote the direction of the spin by a single variable $\sigma_i = \pm 1$ (an Ising spin). The energy of a configuration $\{\sigma_i\}$ of spins is then given by

$$\mathcal{H} = -\frac{1}{2} \sum_{i,j=1}^N J_{ij} \sigma_i \sigma_j - h \sum_i \sigma_i,$$

where h is an external magnetic field, and J_{ij} is the interaction energy between spins at sites i and j .

- (a) For N spins we make the drastic approximation that the interaction between all spins is the same, and $J_{ij} = -J/N$ (the equivalent neighbor model). Show that the energy can now be written as $E(M, h) = -N[Jm^2/2 + hm]$, with a magnetization $m = \sum_{i=1}^N \sigma_i/N = M/N$.
- (b) Show that the partition function $Z(h, T) = \sum_{\{\sigma_i\}} \exp(-\beta \mathcal{H})$ can be rewritten as $Z = \sum_M \exp(-\beta F(m, h))$, with $F(m, h)$ easily calculated by analogy to problem (1). For the remainder of the problem work only with $F(m, h)$ expanded to fourth order in m .
- (c) By saddle point integration show that the actual free energy $F(h, T) = -k_B T \ln Z(h, T)$ is given by $F(h, T) = \min_m \{F(m, h)\}_m$. When is the saddle point method valid? Note that $F(m, h)$ is an analytic function but not convex for $T < T_c$, while the true free energy $F(h, T)$ is convex but becomes non-analytic due to the minimization.
- (d) For $h = 0$ find the critical temperature T_c below which spontaneous magnetization appears; and calculate the magnetization $m(T)$ in the low temperature phase.
- (e) Calculate the singular (non-analytic) behavior of the response functions

$$C = \left. \frac{\partial E}{\partial T} \right|_{h=0} \quad \text{and} \quad \chi = \left. \frac{\partial m}{\partial h} \right|_{h=0}.$$

Solution:

(a)

Problem 1.3

The lattice-gas model: Consider a gas of particles subject to a Hamiltonian

$$\mathcal{H} = \sum_{i=1}^N \frac{\vec{p}_i^2}{2m} + \frac{1}{2} \sum_{i \neq j} \mathcal{V}(\vec{r}_i - \vec{r}_j),$$

in a volume V .

- (a) Show that the grand partition function Ξ can be written as

$$\Xi = \sum_{N=0}^{\infty} \frac{1}{N!} \left(\frac{e^{\beta \mu}}{\lambda^3} \right)^N \int \prod_{i=1}^N d^3 \vec{r}_i \exp \left(-\frac{\beta}{2} \sum_{i \neq j} \mathcal{V}(\vec{r}_i - \vec{r}_j) \right),$$

where λ is the thermal de Broglie wavelength.

- (b) The volume V is now subdivided into $\mathcal{M} = V/a^3$ cells of volume a^3 , with the spacing a chosen small enough so that each cell α is either empty or occupied by one particle; i.e. the cell occupation number n_α is restricted to 0 or 1 ($\alpha = 1, 2, \dots, \mathcal{M}$). After approximating the integrals $\int d^3\vec{r}$ by sums $a^3 \sum_{\alpha=1}^{\mathcal{M}}$, show that

$$\Xi \approx \sum_{\{n_\alpha=0,1\}} \prod_{\alpha} \left(\frac{e^{\beta\mu} a^3}{\lambda^3} \right)^{n_\alpha} \exp \left(-\frac{\beta}{2} \sum_{\alpha,\beta=1}^{\mathcal{M}} n_\alpha n_\beta \mathcal{V}(\vec{r}_\alpha - \vec{r}_\beta) \right).$$

- (c) By setting $n_\alpha = (1 + \sigma_\alpha)/2$ and approximating the potential by $\mathcal{V}(\vec{r}_\alpha - \vec{r}_\beta) = -J/\mathcal{M}$, show that this model is identical to the one studied in problem (2). What does this imply about the behavior of this imperfect gas?

Solution:

(a)

Problem 1.4

Surfactant condensation: N surfactant molecules are added to the surface of water over an area A . They are subject to a Hamiltonian

$$\mathcal{H} = \sum_{i=1}^N \frac{\vec{p}_i^2}{2m} + \frac{1}{2} \sum_{i \neq j} \mathcal{V}(\vec{r}_i - \vec{r}_j),$$

where $\vec{r}_i$ and $\vec{p}_i$ are two-dimensional vectors indicating the position and momentum of particle i .

- Write down the expression for the partition function $Z(N, T, A)$ in terms of integrals over $\vec{r}_i$ and $\vec{p}_i$, and perform the integrals over the momenta. The interparticle potential $\mathcal{V}(\vec{r})$ is infinite for $|\vec{r}| < a$, and attractive for $|\vec{r}| > a$ such that $\int_a^\infty 2\pi r dr \mathcal{V}(r) = -u_0$.
- Estimate the total non-excluded area available in the positional phase space of the system of N particles.
- Estimate the total potential energy of the system, assuming a uniform density $n = N/A$. Using this potential energy for all configurations allowed in the previous part, write down an approximation for Z .
- The surface tension of water without surfactants is σ_0 , approximately independent of temperature. Calculate the surface tension $\sigma(n, T)$ in the presence of surfactants.
- Show that below a certain temperature T_c , the expression for σ is manifestly incorrect. What do you think happens at low temperatures?
- Compute the heat capacities C_A , and write down an expression for C without explicit evaluation, due to the surfactants.

Solution:

(a)

Problem 1.5

Critical behavior of a gas: The pressure P of a gas is related to its density $n = N/V$, and temperature T by the truncated expansion

$$P = k_B T \left(n - \frac{b}{2} n^2 + \frac{c}{6} n^3 \right),$$

where b and c are assumed to be positive, temperature independent constants.

- (a) *Locate the critical temperature T_c below which this equation must be invalid, and the corresponding density n_c and pressure P_c of the critical point. Hence find the ratio $k_B T_c n_c / P_c$.*
- (b) *Calculate the isothermal compressibility $\kappa_T = -\frac{1}{V} \frac{\partial V}{\partial P} \Big|_T$, and sketch its behavior as a function of T for $n = n_c$.*
- (c) *On the critical isotherm give an expression for $P - P_c$ as a function of $n - n_c$.*
- (d) *The instability in the isotherms for $T < T_c$ is avoided by phase separation into a liquid of density n_+ and gas of density n_- . For temperatures close to T_c , these densities behave as $n_{\pm} \approx n_c(1 \pm \varepsilon)$. Using a Maxwell construction, or otherwise, find an implicit equation for $\varepsilon(T)$, and indicate its behavior for $T_c - T \rightarrow 0$. (Hint: Along an isotherm, variations of chemical potential obey $d\mu = dP/n$.)*
- (e) *Now consider a gas obeying Dieterici's equation of state:*

$$P(v - b) = k_B T \exp\left(-\frac{a}{k_B T v}\right),$$

where $v = V/N$. Find the ratio $Pv/k_B T$ at its critical point.

- (f) *Calculate the isothermal compressibility κ_T for $v = v_c$ as a function of $T - T_c$ for the Dieterici gas.*
- (g) *On the Dieterici critical isotherm expand the pressure to the lowest non-zero order in $v - v_c$.*

Solution:

(a)

Problem 1.6

Magnetic thin films: A crystalline film (simple cubic) is obtained by depositing a finite number of layers n . Each atom has a three-component (Heisenberg) spin, and they interact through the Hamiltonian

$$-\beta \mathcal{H} = \sum_{\ell=1}^n \sum_{\langle ij \rangle} J_H \vec{s}_i^{\ell} \cdot \vec{s}_j^{\ell} + \sum_{\ell=1}^{n-1} \sum_i J_V \vec{s}_i^{\ell} \cdot \vec{s}_i^{\ell+1},$$

where the unit vector $\vec{s}_i^{\ell}$ indicates the spin at site i in the ℓ -th layer. A mean-field approximation is obtained from the variational density $\rho_0 \propto \exp(-\beta \mathcal{H}_0)$, with the trial Hamiltonian

$$-\beta \mathcal{H}_0 = \sum_{\ell=1}^n \sum_i \vec{h}^{\ell} \cdot \vec{s}_i^{\ell}.$$

- (a) Calculate the partition function $Z_0(\{\vec{h}^\ell\})$, and $F_0 = -\ln Z_0$.
- (b) Obtain the magnetizations $m^\ell = \langle \vec{s}_i^\ell \rangle_0$, and $\partial^2 F_0 / \partial h_0^2$, in terms of the Langevin function $\mathcal{L}(h) = \coth h - 1/h$.
- (c) Calculate $\langle \mathcal{H} \rangle_0$, with the (reasonable) assumption that all the variational fields $\{\vec{h}^\ell\}$ are parallel.
- (d) The exact free energy $F = -\ln Z$ satisfies the Gibbs inequality $F \leq F_0 + \langle \mathcal{H} - \mathcal{H}_0 \rangle_0$. Give the self-consistent equations for the magnetizations m^ℓ that optimize F_0 . How would you solve these equations numerically?
- (e) Find the critical temperature, and the behavior of the magnetization in the bulk by considering the limit $n \rightarrow \infty$. (Note that $\lim_{m \rightarrow 0} \mathcal{L}^{-1}(m) = 3m + 9m^3/5 + \dots$.)
- (f) By linearizing the self-consistent equations, show that the critical temperature of the film depends on the number of layers n as

$$kT_c(n \gg 1) \approx kT_c - \frac{J_V^2}{3n^2}.$$

- (g) Derive a continuum form of the self-consistent equations, and keep terms to cubic order in m . Show that the resulting nonlinear differential equation has a solution of the form $m(x) = m_{\text{bulk}} \tanh(kx)$. What circumstances are described by this solution?
- (h) How can the above solution be modified to describe a semi-infinite system? Obtain the critical behaviors of the healing length $\xi \sim 1/k$.
- (i) Show that the magnetization of the surface layer vanishes as $|T - T_c|^\beta$.

Solution:

(a)

Chapter 2

Statistical fields

Problem 2.1

Cubic invariants: When the order parameter m goes to zero discontinuously, the phase transition is said to be first order (discontinuous). A common example occurs in systems where symmetry considerations do not exclude a cubic term in the Landau free energy, as in

$$\mathcal{H} = \int d^d x \left[\frac{K}{2} (\nabla m)^2 + \frac{t}{2} m^2 + cm^3 + um^4 \right], \quad K, c, u > 0.$$

- (a) By plotting the energy density $\mathcal{H}(m)$ for uniform m at various values of t , show that as t is reduced there is a discontinuous jump to $m \neq 0$ for a positive t in the saddle point approximation.
- (b) By writing down the two conditions that m and t must satisfy at the transition, solve for m and t .
- (c) Note that the correlation length ξ is related to the curvature of $\mathcal{H}(m)$ at its minimum by $K\xi^{-2} = \partial^2 \mathcal{H} / \partial m^2|_{m_{\text{eq}}}$. Plot ξ as a function of t .

Solution:

(a)

Problem 2.2

Tricritical point: By tuning an additional parameter, a second-order transition can be made first order. The special point separating the two types of transitions is known as a tricritical point, and can be studied by examining the Landau–Ginzburg Hamiltonian

$$\mathcal{H} = \int d^d x \left[\frac{K}{2} (\nabla m)^2 + \frac{t}{2} m^2 + um^4 + vm^6 - hm \right],$$

where u can be positive or negative. For $u < 0$, a positive v is necessary to ensure stability.

- (a) By sketching the energy density $\mathcal{H}(m)$ for various t , show that in the saddle point approximation there is a first-order transition for $u < 0$ and $h = 0$.
- (b) Calculate t and the discontinuity Δm at this transition.

- (c) For $h = 0$ and $v > 0$, plot the phase boundary in the (u, t) plane, identifying the phases, and order of the phase transitions.
- (d) The special point $u = t = 0$, separating first and second-order phase boundaries, is a tricritical point. For $u = 0$, calculate the tricritical exponents α , β , γ , and δ , governing the singularities in magnetization, susceptibility, and heat capacity. (Recall: $C \propto t^{-\alpha}$; $m|_{h=0} \propto t^\beta$; $\chi \propto t^{-\gamma}$; and $m|_{t=0} \propto h^{1/\delta}$.)

Solution:

(a)

Problem 2.3

Transverse susceptibility: An n -component magnetization field $\vec{m}(\vec{x})$ is coupled to an external field $\vec{h}$ through a term $-\int d^d x \vec{h} \cdot \vec{m}(\vec{x})$ in the Hamiltonian $\mathcal{H}$. If $\mathcal{H}$ for $\vec{h} = 0$ is invariant under rotations of $\vec{m}(\vec{x})$; then the free energy density ($f = -\ln Z/V$) only depends on the absolute value of $\vec{h}$; i.e. $f(\vec{h}) = f(h)$, where $h = |\vec{h}|$.

- (a) Show that $\vec{m} = \langle \int d^d x \vec{m}(\vec{x}) / V \rangle = -h f'(h)/h$.
- (b) Relate the susceptibility tensor $\chi_{\alpha\beta} = \partial m_\alpha / \partial h_\beta$, to $f''(h)$, $\vec{m}$, and $\vec{h}$.
- (c) Show that the transverse and longitudinal susceptibilities are given by $\chi_t = m/h$ and $\chi_\ell = -f''(h)$, where m is the magnitude of $\vec{m}$.
- (d) Conclude that χ_t diverges as $\vec{h} \rightarrow 0$, whenever there is a spontaneous magnetization. Is there any similar a priori reason for χ_ℓ to diverge?

Solution:

(a)

Problem 2.4

Superfluid He^4 - He^3 mixtures: The superfluid He^4 order parameter is a complex number $\psi(\vec{x})$. In the presence of a concentration $c(\vec{x})$ of He^3 impurities, the system has the following Landau-Ginzburg energy

$$\mathcal{H}[c, \psi] = \int d^d x \left[\frac{K}{2} |\nabla \psi|^2 + \frac{t}{2} |\psi|^2 + u |\psi|^4 + v |\psi|^6 + \frac{\alpha}{2} (c(\vec{x}) - \bar{c})^2 + \gamma (c(\vec{x}) - \bar{c}) |\psi|^2 \right],$$

with positive K , u , and v .

- (a) Integrate out the He^3 concentrations to find the effective Hamiltonian $\mathcal{H}_{\text{eff}}[\psi]$ for the superfluid order parameter, given by

$$Z = \int \mathcal{D}\psi e^{-\mathcal{H}_{\text{eff}}[\psi]} \equiv \int \mathcal{D}c \mathcal{D}\psi e^{-\mathcal{H}[c, \psi]}.$$

- (b) Obtain the phase diagram for $\mathcal{H}_{\text{eff}}$ using a saddle point approximation. Find the limiting value of γ^* above which the phase transition becomes discontinuous.
- (c) The discontinuous transition is accompanied by a jump in the magnitude of ψ . How does this jump vanish as $\gamma \rightarrow \gamma^*$?
- (d) Show that the discontinuous transition is accompanied by a jump in He^3 concentration.
- (e) Sketch the phase boundary in the (t, γ) coordinates, and indicate how its two segments join at γ^* .
- (f) Going back to the original joint probability for the fields $c(\vec{x})$ and $\psi(\vec{x})$, show that $\langle c(\vec{x}) - \bar{c} \rangle = -(\gamma/\alpha)|\psi(\vec{x})|^2$.
- (g) Show that $\langle c(\vec{x})c(\vec{y}) \rangle = \bar{c}^2 + (\gamma^2/\alpha^2)\langle |\psi(\vec{x})|^2 |\psi(\vec{y})|^2 \rangle$, for $\vec{x} \neq \vec{y}$.
- (h) Qualitatively discuss how $\langle c(\vec{x})c(\vec{0}) \rangle$ decays with $x = |\vec{x}|$ in the disordered phase.
- (i) Qualitatively discuss how $\langle c(\vec{x})c(\vec{0}) \rangle$ decays to its asymptotic value in the ordered phase.

Solution:

(a)

Problem 2.5

Crumpled surfaces: The configurations of a crumpled sheet of paper can be described by a vector field $\vec{r}(\vec{x})$, denoting the position in three-dimensional space, $\vec{r} = (r_1, r_2, r_3)$, of the point at location $\vec{x} = (x_1, x_2)$ on the flat sheet. The energy of each configuration is assumed to be invariant under translations and rotations of the sheet of paper.

- (a) Show that the two lowest-order (in derivatives) terms in the quadratic part of a Landau–Ginzburg Hamiltonian for this system are:

$$\mathcal{H}_0[\vec{r}] = \sum_{\alpha=1,2} \int d^2x \left[\frac{t}{2} (\partial_\alpha \vec{r}) \cdot (\partial_\alpha \vec{r}) + \frac{K}{2} (\nabla^2 \vec{r}) \cdot (\nabla^2 \vec{r}) \right].$$

- (b) Write down the lowest-order terms (there are two) that appear at the quartic level.
- (c) Discuss what happens when t changes sign, assuming that quartic terms provide the required stability (and $K > 0$).

Solution:

(a)

Chapter 3

Fluctuations

Problem 3.1

Spin waves: In the XY model of $n = 2$ magnetism, a unit vector $\vec{s} = (s_x, s_y)$ (with $s_x^2 + s_y^2 = 1$) is placed on each site of a d -dimensional lattice. There is an interaction that tends to keep nearest-neighbors parallel, i.e. a Hamiltonian

$$-\beta\mathcal{H} = K \sum_{\langle ij \rangle} \vec{s}_i \cdot \vec{s}_j.$$

The notation $\langle ij \rangle$ is conventionally used to indicate summing over all nearest-neighbor pairs ij .

- (a) *Rewrite the partition function $Z = \int \prod_i d\vec{s}_i \exp(-\beta\mathcal{H})$, as an integral over the set of angles θ_i between the spins $\vec{s}_i$ and some arbitrary axis.*
- (b) *At low temperatures ($K \gg 1$), the angles θ_i vary slowly from site to site. In this case expand $-\beta\mathcal{H}$ to get a quadratic form in θ_i .*
- (c) *For $d = 1$, consider L sites with periodic boundary conditions (i.e. forming a closed chain). Find the normal modes θ_q that diagonalize the quadratic form (by Fourier transformation), and the corresponding eigenvalues K_q . Pay careful attention to whether the modes are real or complex, and to the allowed values of q .*
- (d) *Generalize the results from the previous part to a d -dimensional simple cubic lattice with periodic boundary conditions.*
- (e) *Calculate the contribution of these modes to the free energy and heat capacity. (Evaluate the classical partition function, i.e. do not quantize the modes.)*
- (f) *Find an expression for $\langle \vec{s}(\vec{0}) \cdot \vec{s}(\vec{x}) \rangle = \text{Re}(\exp[i(\theta_{\vec{x}} - \theta_{\vec{0}})])$ by adding contributions from different Fourier modes. Convince yourself that for $|\vec{x}| \rightarrow \infty$, only $q \rightarrow 0$ modes contribute appreciably to this expression, and hence calculate the asymptotic limit.*
- (g) *Calculate the transverse susceptibility from $\chi_t \propto \int d^d x \langle \vec{s}(\vec{0}) \cdot \vec{s}(\vec{x}) \rangle_c$. How does it depend on the system size L ?*
- (h) *In $d = 2$, show that χ_t only diverges for K larger than a critical value $K_c = 1/(4\pi)$.*

Solution:

(a)

Problem 3.2

Capillary waves: A reasonably flat surface in d dimensions can be described by its height h , as a function of the remaining $d - 1$ coordinates $\vec{x} = (x_1, \dots, x_{d-1})$. Convince yourself that the generalized “area” is given by $\mathcal{A} = \int d^{d-1}x \sqrt{1 + (\nabla h)^2}$. With a surface tension σ , the Hamiltonian is simply $\mathcal{H} = \sigma \mathcal{A}$.

- (a) At sufficiently low temperatures, there are only slow variations in h . Expand the energy to quadratic order, and write down the partition function as a functional integral.
- (b) Use Fourier transformation to diagonalize the quadratic Hamiltonian into its normal modes $\{h_{\vec{q}}\}$ (capillary waves).
- (c) What symmetry breaking is responsible for these Goldstone modes?
- (d) Calculate the height–height correlations $\langle (h(\vec{x}) - h(\vec{x}'))^2 \rangle$.
- (e) Comment on the form of the result in (d) in dimensions $d = 4, 3, 2$, and 1 .
- (f) By estimating typical values of h , comment on when it is justified to ignore higher order terms in the expansion for $\mathcal{A}$.

Solution:

(a)

Problem 3.3

Gauge fluctuations in superconductors: The Landau–Ginzburg model of superconductivity describes a complex superconducting order parameter $\psi(\vec{x}) = \psi_1(\vec{x}) + i\psi_2(\vec{x})$, and the electromagnetic vector potential $\vec{A}(\vec{x})$, which are subject to a Hamiltonian

$$\mathcal{H} = \int d^3x \left[\frac{t}{2} |\psi|^2 + u |\psi|^4 + \frac{K}{2} |D_\mu \psi|^2 + \frac{L}{2} |\nabla \times \vec{A}|^2 \right].$$

The gauge-invariant derivative $D_\mu \equiv \partial_\mu - ieA_\mu(\vec{x})$ introduces the coupling between the two fields. (In terms of Cooper pair parameters, $e = e^*c/\hbar$, $K = \hbar^2/2m^*$.)

- (a) Show that the above Hamiltonian is invariant under the local gauge symmetry:

$$\psi(\vec{x}) \rightarrow \psi(\vec{x}) \exp(i\chi(\vec{x})) \quad \text{and} \quad A_\mu(\vec{x}) \rightarrow A_\mu(\vec{x}) + \frac{1}{e} \partial_\mu \chi.$$

- (b) Show that there is a saddle point solution of the form $\psi(\vec{x}) = \bar{\psi}$ and $\vec{A}(\vec{x}) = 0$, and find $\bar{\psi}$ for $t > 0$ and $t < 0$.
- (c) For $t < 0$, calculate the cost of fluctuations by setting

$$\psi(\vec{x}) = (\bar{\psi} + \phi(\vec{x})) \exp(i\theta(\vec{x})), \quad \vec{A}(\vec{x}) = \vec{a}(\vec{x}),$$

with $\nabla \cdot \vec{a} = 0$ in the Coulomb gauge, and expanding $\mathcal{H}$ to quadratic order in ϕ , θ , and $\vec{a}$.

- (d) Perform a Fourier transformation, and calculate the expectation values $\langle |\phi_{\vec{q}}|^2 \rangle$, $\langle |\theta_{\vec{q}}|^2 \rangle$, and $\langle |\vec{a}_{\vec{q}}|^2 \rangle$.

Solution:

(a)

Problem 3.4

Fluctuations around a tricritical point: As shown in a previous problem, the Hamiltonian

$$\mathcal{H} = \int d^d x \left[\frac{K}{2} (\nabla m)^2 + \frac{t}{2} m^2 + u m^4 + v m^6 \right],$$

with $u = 0$ and $v > 0$ describes a tricritical point.

- (a) Calculate the heat capacity singularity as $t \rightarrow 0$ by the saddle point approximation.
 (b) Include both longitudinal and transverse fluctuations by setting

$$\vec{m}(\vec{x}) = (m + \phi_\ell(\vec{x})) \hat{e}_\ell + \sum_{\alpha=2}^n \phi_{t,\alpha}(\vec{x}) \hat{e}_\alpha,$$

and expanding $\mathcal{H}$ to quadratic order in ϕ .

- (c) Calculate the longitudinal and transverse correlation functions.
 (d) Compute the first correction to the saddle point free energy from fluctuations.
 (e) Find the fluctuation correction to the heat capacity.
 (f) By comparing the results from parts (a) and (e) for $t < 0$, obtain a Ginzburg criterion, and the upper critical dimension for validity of mean-field theory at a tricritical point.
 (g) A generalized multicritical point is described by replacing the term vm^6 with $u_{2n}m^{2n}$. Use simple power counting to find the upper critical dimension of this multicritical point.

Solution:

(a)

Problem 3.5

Coupling to a “massless” field: Consider an n -component vector field $\vec{m}(\vec{x})$ coupled to a scalar field $A(\vec{x})$, through the effective Hamiltonian

$$\mathcal{H} = \int d^d x \left[\frac{K}{2} (\nabla \vec{m})^2 + \frac{t}{2} \vec{m}^2 + u (\vec{m}^2)^2 + e^2 \vec{m}^2 A^2 + \frac{L}{2} (\nabla A)^2 \right],$$

with K , L , and u positive.

- (a) Show that there is a saddle point solution of the form $\vec{m}(\vec{x}) = m\hat{e}_\ell$ and $A(\vec{x}) = 0$, and find m for $t > 0$ and $t < 0$.
- (b) Sketch the heat capacity $C = \partial^2 \ln Z / \partial t^2$, and discuss its singularity as $t \rightarrow 0$ in the saddle point approximation.
- (c) Include fluctuations by setting $\vec{m}(\vec{x}) = (m + \phi_\ell(\vec{x}))\hat{e}_\ell + \phi_t(\vec{x})\hat{e}_t$ and $A(\vec{x}) = a(\vec{x})$, and expand $\mathcal{H}$ to quadratic order in ϕ and a .
- (d) Find the correlation lengths ξ_ℓ and ξ_t for the longitudinal and transverse components of ϕ , for $t > 0$ and $t < 0$.
- (e) Find the correlation length ξ_a for the fluctuations of the scalar field a , for $t > 0$ and $t < 0$.
- (f) Calculate the correlation function $\langle a(\vec{x})a(\vec{0}) \rangle$ for $t > 0$.
- (g) Compute the correction to the saddle point free energy $\ln Z$, from fluctuations. (You can leave the answer in the form of integrals involving ξ_ℓ , ξ_t , and ξ_a .)
- (h) Find the fluctuation corrections to the heat capacity in (b), again leaving the answer in the form of integrals.
- (i) Discuss the behavior of the integrals appearing above schematically, and state their dependence on the correlation length ξ , and cutoff Λ , in different dimensions.
- (j) What is the critical dimension for the validity of saddle point results, and how is it modified by the coupling to the scalar field?

Solution:

(a)

Problem 3.6

Random magnetic fields: Consider the Hamiltonian

$$\mathcal{H} = \int d^d x \left[\frac{K}{2} (\nabla m)^2 + \frac{t}{2} m^2 + u m^4 - h(\vec{x}) m(\vec{x}) \right],$$

where $m(\vec{x})$ and $h(\vec{x})$ are scalar fields, and $u > 0$. The random magnetic field $h(\vec{x})$ results from frozen (quenched) impurities that are independently distributed in space. For simplicity $h(\vec{x})$ is assumed to be an independent Gaussian variable at each point $\vec{x}$, such that

$$\overline{h(\vec{x})} = 0 \quad \text{and} \quad \overline{h(\vec{x})h(\vec{x}')} = \Delta \delta^d(\vec{x} - \vec{x}'),$$

where the overline indicates (quench) averaging over all values of the random fields. The above equation implies that the Fourier transformed random field $\tilde{h}(\vec{q})$ satisfies

$$\overline{\tilde{h}(\vec{q})} = 0 \quad \text{and} \quad \overline{\tilde{h}(\vec{q})\tilde{h}(\vec{q}')} = \Delta (2\pi)^d \delta^d(\vec{q} + \vec{q}').$$

- (a) Calculate the quench averaged free energy $f_{\text{sp}} = \min_m \mathcal{H}(m)$, assuming a saddle point solution with uniform magnetization $m(\vec{x}) = m$.
- (b) Include fluctuations by setting $m(\vec{x}) = m + \phi(\vec{x})$, and expand $\mathcal{H}$ to second order in ϕ .

- (c) Express the energy cost of the above fluctuations in terms of the Fourier modes $\phi_{\vec{q}}$.
- (d) Calculate the mean $\langle \phi_{\vec{q}} \rangle$ and the variance $\langle |\phi_{\vec{q}}|^2 \rangle_c$, where $\langle \dots \rangle$ denotes the usual thermal expectation value for a fixed $\tilde{h}(\vec{q})$.
- (e) Use the above results, in conjunction with the quench average, to calculate the scattering line shape $S(\vec{q}) = \overline{\langle |\phi_{\vec{q}}|^2 \rangle}$.
- (f) Perform the Gaussian integrals over $\phi_{\vec{q}}$ to calculate the fluctuation corrections $f[\tilde{h}(\vec{q})]$ to the free energy.
- (g) Use the quench average to calculate the corrections due to the fluctuations in the previous part to the quench averaged free energy $\bar{f}$. (Leave the corrections in the form of two integrals.)
- (h) Estimate the singular t dependence of the integrals obtained in the fluctuation corrections to the free energy.
- (i) Find the upper critical dimension d_u for the validity of saddle point critical behavior.

Solution:

(a)

Problem 3.7

Long-range interactions: Consider a continuous spin field $\vec{s}(\vec{x})$, subject to a long-range ferromagnetic interaction

$$\int d^d x d^d y \frac{\vec{s}(\vec{x}) \cdot \vec{s}(\vec{y})}{|\vec{x} - \vec{y}|^{d+\sigma}},$$

as well as short-range interactions.

- (a) How is the quadratic term in the Landau–Ginzburg expansion modified by the presence of this long-range interaction? For what values of σ is the long-range interaction dominant?
- (b) By estimating the magnitude of thermally excited Goldstone modes (or otherwise), obtain the lower critical dimension d_l below which there is no long-range order.
- (c) Find the upper critical dimension d_u , above which saddle point results provide a correct description of the phase transition.

Solution:

(a)

Problem 3.8

Ginzburg criterion along the magnetic field direction: Consider the Hamiltonian

$$\mathcal{H} = \int d^d x \left[\frac{K}{2} (\nabla \vec{m})^2 + \frac{t}{2} \vec{m}^2 + u (\vec{m}^2)^2 - \vec{h} \cdot \vec{m} \right],$$

describing an n -component magnetization vector $\vec{m}(\vec{x})$, with $u > 0$.

- (a) In the saddle point approximation, the free energy is $f = \min_m \mathcal{H}(m)$. Indicate the resulting phase boundary in the (h, t) plane, and label the phases. (h denotes the magnitude of $\vec{h}$.)
- (b) Sketch the form of m for $t < 0$, on both sides of the phase boundary, and for $t > 0$ at $h = 0$.
- (c) For t and h close to zero, the spontaneous magnetization can be written as $m = t^\beta g_m(h/t^\Delta)$. Identify the exponents β and Δ in the saddle point approximation. For the remainder of this problem set $t = 0$.
- (d) Calculate the transverse and longitudinal susceptibilities at a finite h .
- (e) Include fluctuations by setting $\vec{m}(\vec{x}) = (m + \phi_\ell(\vec{x}))\hat{e}_\ell + \vec{\phi}_t(\vec{x})\hat{e}_t$ and expanding $\mathcal{H}$ to second order in the ϕ s. ($\hat{e}_\ell$ is a unit vector parallel to the average magnetization, and $\hat{e}_t$ is perpendicular to it.)
- (f) Calculate the longitudinal and transverse correlation lengths.
- (g) Calculate the first correction to the free energy from these fluctuations. (The scaling form is sufficient.)
- (h) Calculate the first correction to the magnetization, and to the longitudinal susceptibility, from the fluctuations.
- (i) By comparing the saddle point value with the correction due to fluctuations, find the upper critical dimension d_u for the validity of the saddle point result.
- (j) For $d < d_u$ obtain a Ginzburg criterion by finding the field h_G below which fluctuations are important. (You may ignore the numerical coefficients in h_G , but the dependences on K and u are required.)

Solution:

(a)

Chapter 4

The scaling hypothesis

Problem 4.1

Scaling in fluids: Near the liquid–gas critical point, the free energy is assumed to take the scaling form $F/N = t^{2-\alpha}g(\Delta\rho/t^\beta)$, where $t = (T - T_c)/T_c$ is the reduced temperature, and $\Delta\rho = \rho - \rho_c$ measures deviations from the critical point density. The leading singular behavior of any thermodynamic parameter $Q(t)$ is of the form t^x on approaching the critical point along the isochore $\rho = \rho_c$; or $(\Delta\rho)^y$ for a path along the isotherm $T = T_c$. Find the exponents x and y for the following quantities:

- (a) The internal energy per particle $\langle\mathcal{H}\rangle/N$, and the entropy per particle $s = S/N$.
- (b) The heat capacities $C_V = T(\partial s/\partial T)_V$ and $C_P = T(\partial s/\partial T)_P$.
- (c) The isothermal compressibility $\kappa_T = (\partial\rho/\partial P)_T/\rho$, and the thermal expansion coefficient $\alpha_P = (\partial V/\partial T)_P/V$. Check that your results for parts (b) and (c) are consistent with the thermodynamic identity $C_P - C_V = TV\alpha_P^2/\kappa_T$.
- (d) Sketch the behavior of the latent heat per particle L on the coexistence curve for $T < T_c$, and find its singularity as a function of t .

Solution:

(a)

Problem 4.2

The Ising model: The differential recursion relations for temperature T , and magnetic field h , of the Ising model in $d = 1 + \epsilon$ dimensions are (for $b = e^\ell$)

$$\begin{cases} \frac{dT}{d\ell} = -\epsilon T + \frac{1}{2}T^2, \\ \frac{dh}{d\ell} = d h. \end{cases}$$

- (a) Sketch the renormalization group flows in the (T, h) plane (for $\epsilon > 0$), marking the fixed points along the $h = 0$ axis.
- (b) Calculate the eigenvalues y_t and y_h , at the critical fixed point, to order of ϵ .

- (c) Starting from the relation governing the change of the correlation length ξ under renormalization, show that $\xi(t, h) = t^{-\nu} g(h/t^\Delta)$ (where $t = T/T_c - 1$), and find the exponents ν and Δ .
- (d) Use a hyperscaling relation to find the singular part of the free energy $f_{\text{sing}}(t, h)$, and hence the heat capacity exponent α .
- (e) Find the exponents β and γ for the singular behaviors of the magnetization and susceptibility, respectively.
- (f) Starting with the relation between susceptibility and correlations of local magnetizations, calculate the exponent η for the critical correlations ($\langle m(\vec{0})m(\vec{x}) \rangle \sim |\vec{x}|^{-(d-2+\eta)}$).
- (g) How does the correlation length diverge as $T \rightarrow 0$ (along $h = 0$) for $d = 1$?

Solution:

(a)

Problem 4.3

The nonlinear σ model describes n -component unit spins. As we shall demonstrate later, in $d = 2$ dimensions, the recursion relations for temperature T , and magnetic field h , are (for $b = e^\ell$)

$$\begin{cases} \frac{dT}{d\ell} = \frac{n-2}{2\pi} T^2, \\ \frac{dh}{d\ell} = 2h. \end{cases}$$

- (a) How does the correlation length diverge as $T \rightarrow 0$?
- (b) Write down the singular form of the free energy as $T, h \rightarrow 0$.
- (c) How does the susceptibility χ diverge as $T \rightarrow 0$ for $h = 0$?

Solution:

(a)

Problem 4.4

Coupled scalars: Consider the Hamiltonian

$$\mathcal{H} = \int d^d x \left[\frac{t}{2} m^2 + \frac{K}{2} (\nabla m)^2 - hm + \frac{L}{2} (\nabla \phi)^2 + vm\phi \right],$$

coupling two one-component fields m and ϕ .

- (a) Write $\mathcal{H}$ in terms of the Fourier transforms $m_{\vec{q}}$ and $\phi_{\vec{q}}$.

- (b) Construct a renormalization group transformation as in the text, by rescaling distances such that $\vec{q}' = b\vec{q}$; and the fields such that $m'_{\vec{q}} = \tilde{m}_{\vec{q}}/z$ and $\phi'_{\vec{q}} = \tilde{\phi}_{\vec{q}}/y$. Do not evaluate the integrals that just contribute a constant additive term.
- (c) There is a fixed point such that $K' = K$ and $L' = L$. Find y_t , y_h , and y_v at this fixed point.
- (d) The singular part of the free energy has a scaling form $f_{\text{sing}}(t, h, v) = t^{2-\alpha} g(h/t^\Delta, v/t^\phi)$ for t, h, v close to zero. Find α and ϕ .
- (e) There is another fixed point such that $t' = t$ and $L' = L$. What are the relevant operators at this fixed point, and how do they scale?

Solution:

(a)

Chapter 5

Perturbative renormalization group

Problem 5.1

Longitudinal susceptibility: While there is no reason for the longitudinal susceptibility to diverge at the mean-field level, it in fact does so due to fluctuations in dimensions $d < 4$. This problem is intended to show you the origin of this divergence in perturbation theory.

Consider the Landau–Ginzburg Hamiltonian

$$\mathcal{H} = \int d^d x \left[\frac{K}{2} (\nabla \vec{m})^2 + \frac{t}{2} \vec{m}^2 + u (\vec{m}^2)^2 \right],$$

describing an n -component magnetization vector $\vec{m}(\vec{x})$, in the ordered phase for $t < 0$.

- (a) Let $\vec{m}(\vec{x}) = (m + \phi_\ell(\vec{x}))\hat{e}_\ell + \vec{\phi}_t(\vec{x})\hat{e}_t$, and expand $\mathcal{H}$ keeping all terms in the expansion.
- (b) Regard the quadratic terms in ϕ_ℓ and $\vec{\phi}_t$ as an unperturbed Hamiltonian $\mathcal{H}_0$, and the lowest order term coupling ϕ_ℓ and $\vec{\phi}_t$ as a perturbation U ; i.e.

$$U = 4um \int d^d x \phi_\ell(\vec{x}) |\vec{\phi}_t(\vec{x})|^2.$$

Write U in Fourier space in terms of $\phi_{\ell,\vec{q}}$ and $\vec{\phi}_{t,\vec{q}}$.

- (c) Calculate the Gaussian (bare) expectation values $\langle \phi_{\ell,\vec{q}} \phi_{\ell,\vec{q}'} \rangle_0$ and $\langle \phi_{t,\alpha,\vec{q}} \phi_{t,\beta,\vec{q}'} \rangle_0$, and the corresponding momentum-dependent susceptibilities $\chi_\ell(\vec{q})_0$ and $\chi_t(\vec{q})_0$.
- (d) Calculate $\langle \vec{\phi}_{t,\vec{q}_1} \cdot \vec{\phi}_{t,\vec{q}_2} \vec{\phi}_{t,\vec{q}_1} \cdot \vec{\phi}_{t,\vec{q}_2} \rangle_0$ using Wick's theorem. (Don't forget that $\vec{\phi}_t$ is an $(n-1)$ -component vector.)
- (e) Write down the expression for $\langle \phi_{\ell,\vec{q}} \phi_{\ell,\vec{q}'} \rangle$ to second order in the perturbation U . Note that since U is odd in ϕ_ℓ , only two terms at second order are non-zero.
- (f) Using the form of U in Fourier space, write the correction term as a product of two four-point expectation values similar to those of part (d). Note that only connected terms for the longitudinal four-point function should be included.
- (g) Ignore the disconnected term obtained in (d) (i.e. the part proportional to $(n-1)^2$), and write down the expression for $\chi_\ell(\vec{q})$ in second-order perturbation theory.
- (h) Show that for $d < 4$, the correction term diverges as q^{d-4} for $q \rightarrow 0$, implying an infinite longitudinal susceptibility.

Solution:

(a)

Problem 5.2

Crystal anisotropy: Consider a ferromagnet with a tetragonal crystal structure. Coupling of the spins to the underlying lattice may destroy their full rotational symmetry. The resulting anisotropies can be described by modifying the Landau–Ginzburg Hamiltonian to

$$\mathcal{H} = \int d^d x \left[\frac{K}{2} (\nabla \vec{m})^2 + \frac{t}{2} \vec{m}^2 + u (\vec{m}^2)^2 + \frac{r}{2} m_1^2 + v m_1^2 \vec{m}^2 \right],$$

where $\vec{m} \equiv (m_1, \dots, m_n)$, and $\vec{m}^2 = \sum_{i=1}^n m_i^2$ ($d = n = 3$ for magnets in three dimensions). Here $u > 0$, and to simplify calculations we shall set $v = 0$ throughout.

- (a) For a fixed magnitude $|\vec{m}|$, what directions in the n -component magnetization space are selected for $r > 0$, and for $r < 0$?
- (b) Using the saddle point approximation, calculate the free energies $(\ln Z)$ for phases uniformly magnetized parallel and perpendicular to direction 1.
- (c) Sketch the phase diagram in the (t, r) plane, and indicate the phases (type of order), and the nature of the phase transitions (continuous or discontinuous).
- (d) Are there Goldstone modes in the ordered phases?
- (e) For $u = 0$, and positive t and r , calculate the unperturbed averages $\langle m_{1, \vec{q}} m_{1, -\vec{q}} \rangle_0$ and $\langle m_{2, \vec{q}} m_{2, -\vec{q}} \rangle_0$, where $m_{i, \vec{q}}$ indicates the Fourier transform of $m_i(\vec{x})$.
- (f) Write the fourth-order term $\mathcal{U} \equiv u \int d^d x (\vec{m}^2)^2$, in terms of the Fourier modes $m_{i, \vec{q}}$.
- (g) Treating $\mathcal{U}$ as a perturbation, calculate the first-order correction to $\langle m_{1, \vec{q}} m_{1, -\vec{q}} \rangle$. (You can leave your answers in the form of some integrals.)
- (h) Treating $\mathcal{U}$ as a perturbation, calculate the first-order correction to $\langle m_{2, \vec{q}} m_{2, -\vec{q}} \rangle$.
- (i) Using the above answer, identify the inverse susceptibility χ_{22}^{-1} , and then find the transition point t_c from its vanishing, to first order in u .
- (j) Is the critical behavior different from the isotropic $O(n)$ model in $d < 4$? In RG language, is the parameter r relevant at the $O(n)$ fixed point? In either case indicate the universality classes expected for the transitions.

Solution:

(a)

Problem 5.3

Cubic anisotropy – mean-field treatment: Consider the modified Landau–Ginzburg Hamiltonian

$$\mathcal{H} = \int d^d x \left[\frac{K}{2} (\nabla \vec{m})^2 + \frac{t}{2} \vec{m}^2 + u (\vec{m}^2)^2 + v \sum_{i=1}^n m_i^4 \right],$$

for an n -component vector $\vec{m}(\vec{x}) = (m_1, m_2, \dots, m_n)$. The “cubic anisotropy” term $+v \sum_{i=1}^n m_i^4$ breaks the full rotational symmetry and selects specific directions.

- (a) For a fixed magnitude $|\vec{m}|$, what directions in the n -component magnetization space are selected for $v > 0$ and for $v < 0$? What is the degeneracy of easy magnetization axes in each case?
- (b) What are the restrictions on u and v for $\mathcal{H}$ to have finite minima? Sketch these regions of stability in the (u, v) plane.
- (c) In general, higher-order terms (e.g. $u_6(\vec{m}^2)^3$ with $u_6 > 0$) are present and ensure stability in the regions not allowed in part (b) (as in the case of the tricritical point discussed in earlier problems). With such terms in mind, sketch the saddle-point phase diagram in the (t, v) plane for $u > 0$; clearly identifying the phases, and order of the transition lines.
- (d) Are there any Goldstone modes in the ordered phases?

Solution:

(a)

Problem 5.4

Cubic anisotropy – ϵ -expansion:

- (a) By looking at diagrams in a second-order perturbation expansion in both u and v , show that the recursion relations for these couplings are

$$\begin{cases} \frac{du}{d\ell} = \epsilon u - 4C[(n+8)u^2 + 6uv], \\ \frac{dv}{d\ell} = \epsilon v - 4C[12uv + 9v^2], \end{cases}$$

where $C = K_d \Lambda^d / (t + K \Lambda^2) \approx K_4 / K^2$ is approximately a constant.

- (b) Find all fixed points in the (u, v) plane, and draw the flow patterns for $n < 4$ and $n > 4$. Discuss the relevance of the cubic anisotropy term near the stable fixed point in each case.
- (c) Find the recursion relation for the reduced temperature t , and calculate the exponent ν at the stable fixed points for $n < 4$ and $n > 4$.
- (d) Is the region of stability in the (u, v) plane calculated in part (b) of the previous problem enhanced or diminished by inclusion of fluctuations? Since in reality higher-order terms will be present, what does this imply about the nature of the phase transition for a small negative v and $n > 4$?
- (e) Draw schematic phase diagrams in the (t, v) plane ($u > 0$) for $n > 4$ and $n < 4$, identifying the ordered phases. Are there Goldstone modes in any of these phases close to the phase transition?

Solution:

(a)

Problem 5.5

Exponents: Two critical exponents at second order are

$$\nu = \frac{1}{2} + \frac{n+2}{4(n+8)}\epsilon + \frac{(n+2)(n^2+23n+60)}{8(n+8)^3}\epsilon^2, \quad \eta = \frac{n+2}{2(n+8)^2}\epsilon^2.$$

Use scaling relations to obtain ϵ -expansions for two or more of the remaining exponents α , β , γ , δ , and ν . Make a table of the results obtained by setting $\epsilon = 1, 2$ for $n = 1, 2$, and 3 ; and compare to the best estimates of these exponents that you can find by other sources (series, experiments, etc.).

Solution:

Problem 5.6

Anisotropic criticality: A number of materials, such as liquid crystals, are anisotropic and behave differently along distinct directions, which shall be denoted parallel and perpendicular, respectively. Let us assume that the d spatial dimensions are grouped into $n_{\parallel}$ parallel directions $\vec{x}_{\parallel}$, and $d - n_{\parallel}$ perpendicular directions $\vec{x}_{\perp}$. Consider a one-component field $m(\vec{x}_{\parallel}, \vec{x}_{\perp})$ subject to a Landau–Ginzburg Hamiltonian $\mathcal{H} = \mathcal{H}_0 + \mathcal{U}$, with

$$\mathcal{H}_0 = \int d^{n_{\parallel}}x_{\parallel} d^{d-n_{\parallel}}x_{\perp} \left[\frac{K}{2}(\nabla_{\parallel}m)^2 + \frac{L}{2}(\nabla_{\perp}^2m)^2 + \frac{t}{2}m^2 - hm \right],$$

and $\mathcal{U} = u \int d^{n_{\parallel}}x_{\parallel} d^{d-n_{\parallel}}x_{\perp} m^4$. (Note that $\mathcal{H}$ depends on the first gradient in the $\vec{x}_{\parallel}$ directions, and on the second gradient in the $\vec{x}_{\perp}$ directions.)

- (a) Write $\mathcal{H}_0$ in terms of the Fourier transforms $m(\vec{q}_{\parallel}, \vec{q}_{\perp})$.
- (b) Construct a renormalization group transformation for $\mathcal{H}_0$, by rescaling coordinates such that $\vec{q}_{\parallel}' = b\vec{q}_{\parallel}$ and $\vec{q}_{\perp}' = c\vec{q}_{\perp}$, and the field as $m'_{\vec{q}} = m_{\vec{q}}/z$. Note that parallel and perpendicular directions are scaled differently. Write down the recursion relations for K , L , t , and h in terms of b , c , and z .
- (c) Choose $c(b)$ and $z(b)$ such that $K' = K$ and $L' = L$. At the resulting fixed point calculate the eigenvalues y_t and y_h for the rescalings of t and h .
- (d) Write the relationship between the (singular parts of) free energies $f(t, h)$ and $f'(t', h')$ in the original and rescaled problems. Hence write the unperturbed free energy in the homogeneous form $f(t, h) = t^{2-\alpha}g_f(h/t^{\Delta})$, and identify the exponents α and Δ .
- (e) How does the unperturbed zero-field susceptibility $\chi(t, h=0)$ diverge as $t \rightarrow 0$? In the remainder of this problem set $h=0$, and treat $\mathcal{U}$ as a perturbation.
- (f) In the unperturbed Hamiltonian calculate the expectation value $\langle m_{\vec{q}}m_{\vec{q}} \rangle_0$, and the corresponding susceptibility $\chi_0(\vec{q}) = \langle |m_{\vec{q}}|^2 \rangle_0$, where $\vec{q}$ stands for $(\vec{q}_{\parallel}, \vec{q}_{\perp})$.
- (g) Write the perturbation $\mathcal{U}$, in terms of the normal modes $m_{\vec{q}}$.
- (h) Using RG, or any other method, find the upper critical dimension d_u for validity of the Gaussian exponents.

- (i) Write down the expansion for $\langle m_{\vec{q}} m_{-\vec{q}} \rangle$, to first order in $\mathcal{U}$, and reduce the correction term to a product of two-point expectation values.
- (j) Write down the expression for $\chi(\vec{q})$, in first-order perturbation theory, and identify the transition point t_c at order of u . (Do not evaluate any integrals explicitly.)

Solution:

(a)

Problem 5.7

Long-range interactions between spins can be described by adding a term

$$\int d^d x d^d y J(\vec{x} - \vec{y}) \vec{m}(\vec{x}) \cdot \vec{m}(\vec{y})$$

to the usual Landau–Ginzburg Hamiltonian.

- (a) Show that for $J(r) \propto 1/r^{d+\sigma}$, the Hamiltonian can be written as

$$\mathcal{H} = \int \frac{d^d q}{(2\pi)^d} \frac{t + K_2 q^2 + K_\sigma q^\sigma + \cdots}{2} \vec{m}(\vec{q}) \cdot \vec{m}(-\vec{q}) + u \int \frac{d^d q_1 d^d q_2 d^d q_3}{(2\pi)^{3d}} [\vec{m}(\vec{q}_1) \cdot \vec{m}(\vec{q}_2)] [\vec{m}(\vec{q}_3) \cdot \vec{m}(-\vec{q}_1 - \vec{q}_2 - \vec{q}_3)]$$

- (b) For $u = 0$, construct the recursion relations for t , K_2 , K_σ and show that K_2 is irrelevant for $\sigma > 2$. What is the fixed Hamiltonian in this case?
- (c) For $\sigma < 2$ and $u = 0$, show that the spin rescaling factor must be chosen such that $K'_\sigma = K_\sigma$, in which case K_2 is irrelevant. What is the fixed Hamiltonian now?
- (d) For $\sigma < 2$, calculate the generalized Gaussian exponents α , β , γ from the recursion relations. Show that u is irrelevant, and hence the Gaussian results are valid, for $d > 2\sigma$.
- (e) For $\sigma < 2$, use a perturbation expansion in u to construct the recursion relations for t , K_σ , u as in the text.
- (f) For $d < 2\sigma$, calculate the critical exponents ν and η to first order in $\epsilon = d - 2\sigma$.
- (g) What is the critical behavior if $J(r) \propto \exp(-r/a)$? Explain!

Solution:

(a)

Chapter 6

Lattice systems

Problem 6.1

Cumulant method: Apply the Niemeijer–van Leeuwen first-order cumulant expansion to the Ising model on a square lattice with $-\beta\mathcal{H} = K \sum_{\langle ij \rangle} \sigma_i \sigma_j$, by following these steps:

- (a) For an RG with $b = 2$, divide the bonds into intracell components $\mathcal{H}_0$, and intercell components $\mathcal{U}$.
- (b) For each cell α , define a renormalized spin $\sigma'_\alpha = \text{sign}(\sigma_1^\alpha + \sigma_2^\alpha + \sigma_3^\alpha + \sigma_4^\alpha)$. This choice becomes ambiguous for configurations such that $\sum_{i=1}^4 \sigma_i^\alpha = 0$. Distribute the weight of these configurations equally between $\sigma'_\alpha = +1$ and -1 (i.e. put a factor of $1/2$ in addition to the Boltzmann weight). Make a table for all possible configurations of a cell, the internal probability $\exp(-\beta\mathcal{H}_0)$, and the weights contributing to $\sigma'_\alpha = \pm 1$.
- (c) Express $\langle \mathcal{U} \rangle_0$ in terms of the cell spins σ'_α , and hence obtain the recursion relation $K'(K)$.
- (d) Find the fixed point K^* , and the thermal eigenvalue y_t .
- (e) In the presence of a small magnetic field $h \sum_i \sigma_i$, find the recursion relation for h , and calculate the magnetic eigenvalue y_h at the fixed point.
- (f) Compare K^* , y_t , and y_h to their exact values.

Solution:

(a)

Problem 6.2

Migdal–Kadanoff method: Consider Potts spins $s_i = 1, 2, \dots, q$, on sites i of a hypercubic lattice, interacting with their nearest neighbors via a Hamiltonian

$$-\beta\mathcal{H} = K \sum_{\langle ij \rangle} \delta_{s_i s_j}.$$

- (a) In $d = 1$ find the exact recursion relations by a $b = 2$ renormalization/decimation process. Identify all fixed points and note their stability.
- (b) Write down the recursion relation $K'(K)$ in d -dimensions for $b = 2$, using the Migdal–Kadanoff bond moving scheme.

- (c) By considering the stability of the fixed points at zero and infinite coupling, prove the existence of a non-trivial fixed point at finite K^* for $d > 1$.
- (d) For $d = 2$, obtain K^* and y_t , for $q = 3, 1$, and 0 .

Solution:

(a)

Problem 6.3

The Potts model: The transfer matrix procedure can be extended to the Potts models, where the spin s_i on each site takes q values $s_i = 1, 2, \dots, q$; and the Hamiltonian is $-\beta\mathcal{H} = K \sum_{i=1}^N \delta_{s_i s_{i+1}} + K \delta_{s_N s_1}$.

- (a) Write down the transfer matrix and diagonalize it. Note that you do not have to solve a q th order secular equation as it is easy to guess the eigenvectors from the symmetry of the matrix.
- (b) Calculate the free energy per site.
- (c) Give the expression for the correlation length ξ (you don't need to provide a detailed derivation), and discuss its behavior as $T = 1/K \rightarrow 0$.

Solution:

(a)

Problem 6.4

The spin-1 model: Consider a linear chain where the spin s_i at each site takes on three values $s_i = -1, 0, +1$. The spins interact via a Hamiltonian

$$-\beta\mathcal{H} = K \sum_i s_i s_{i+1}.$$

- (a) Write down the transfer matrix $\langle s|T|s' \rangle = e^{K s s'}$ explicitly.
- (b) Use symmetry properties to find the largest eigenvalue of T and hence obtain the expression for the free energy per site ($\ln Z/N$).
- (c) Obtain the expression for the correlation length ξ , and note its behavior as $K \rightarrow \infty$.
- (d) If we try to perform a renormalization group by decimation on the above chain we find that additional interactions are generated. Write down the simplest generalization of $-\beta\mathcal{H}$ whose parameter space is closed under such RG.

Solution:

(a)

Problem 6.5

Clock model: Each site of the lattice is occupied by a q -valued spin $s_i \equiv 1, 2, \dots, q$, with an underlying translational symmetry modulus q , i.e. the system is invariant under $s_i \rightarrow s_i + n \pmod{q}$. The most general Hamiltonian subject to this symmetry with nearest-neighbor interactions is

$$\mathcal{H}_C = - \sum_{\langle ij \rangle} J(s_i - s_j \pmod{q}),$$

where $J(n)$ is any function, e.g. $J(n) = J \cos(2\pi n/q)$. Potts models are a special case of clock models with full permutation symmetry, and the Ising model is obtained in the limit of $q = 2$.

- (a) For a closed linear chain of N clock spins subject to the above Hamiltonian, show that the partition function $Z = \text{tr} \exp(-\beta \mathcal{H}_C)$ can be written as

$$Z = \text{tr}[T(s_1, s_2)T(s_2, s_3) \cdots T(s_N, s_1)],$$

where $T \equiv \langle s_i | T | s_j \rangle = \exp(J(s_i - s_j))$ is a $q \times q$ transfer matrix.

- (b) Write down the transfer matrix explicitly and diagonalize it. Note that because of the translational symmetry, the eigenvalues are easily obtained by discrete Fourier transformation as

$$\lambda_k = \sum_{n=1}^q \exp\left(J(n) + \frac{2\pi i n k}{q}\right).$$

- (c) Show that $Z = \sum_{k=1}^q \lambda_k^N \approx \lambda_0^N$ for $N \rightarrow \infty$. Write down the expression for the free energy per site $f = -\ln Z/N$.

- (d) Show that the correlation function can be calculated from

$$\langle s_i s_{i+\ell} \rangle = \frac{1}{Z} \sum_{\alpha} \text{tr}[P_{\alpha} T^{\ell} T^{N-\ell}],$$

where P_{α} is a projection matrix. Hence show that $\langle s_i s_{i+\ell} \rangle_c \sim (\lambda_1/\lambda_0)^{\ell}$.

Solution:

- (a)
-

Problem 6.6

XY model: Consider two-component unit spins $\vec{s}_i = (\cos \theta_i, \sin \theta_i)$ in one dimension, with the nearest-neighbor interactions described by $-\beta \mathcal{H} = K \sum_{i=1}^N \vec{s}_i \cdot \vec{s}_{i+1}$.

- (a) Write down the transfer matrix T' , and show that it can be diagonalized with eigenvectors $f_m \propto e^{im\theta}$ for integer m .
- (b) Calculate the free energy per site, and comment on the behavior of the heat capacity as $T \propto K^{-1} \rightarrow 0$.
- (c) Find the correlation length ξ , and note its behavior as $K \rightarrow \infty$.

Solution:

(a)

Problem 6.7

One-dimensional gas: The transfer matrix method can also be applied to a one-dimensional gas of particles with short-range interactions, as described in this problem.

- (a) *Show that for a potential with a hard core that screens the interactions from further neighbors, the Hamiltonian for N particles can be written as*

$$\mathcal{H} = \sum_{i=1}^N \frac{p_i^2}{2m} + \sum_{i=2}^N V(x_i - x_{i-1}).$$

The (indistinguishable) particles are labeled with coordinates x_i such that $0 \leq x_1 \leq x_2 \leq \dots \leq x_N \leq L$, where L is the length of the box confining the particles.

- (b) *Write the expression for the partition function $Z(T, N, L)$. Change variables to $\xi_1 = x_1$, $\xi_2 = x_2 - x_1$, $\dots$, $\xi_N = x_N - x_{N-1}$, and carefully indicate the allowed ranges of integration and the constraints.*

- (c) *Consider the Gibbs partition function obtained from the Laplace transformation*

$$\mathcal{Z}(T, N, P) = \int_0^\infty dL e^{-\beta PL} Z(T, N, L),$$

and by extremizing the integrand find the standard formula for P in the canonical ensemble.

- (d) *Change variables from L to $\xi_{N+1} = L - \sum_{i=1}^N \xi_i$, and find the expression for $\mathcal{Z}(T, N, P)$ as a product over one-dimensional integrals over each ξ_i .*
- (e) *At a fixed pressure P , find expressions for the mean length $\langle L \rangle(T, N, P)$, and the density $n = N/\langle L \rangle(T, N, P)$ (involving ratios of integrals which should be easy to interpret).*
- (f) *For a hard sphere gas, with minimum separation a between particles, calculate the equation of state $P(T, n)$.*

Solution:

(a)

Problem 6.8

Potts chain (RG): Consider a one-dimensional array of N Potts spins $s_i = 1, 2, \dots, q$, subject to the Hamiltonian $-\beta\mathcal{H} = J \sum_i \delta_{s_i s_{i+1}}$.

- (a) *Using the transfer matrix method (or otherwise) calculate the partition function Z , and the correlation length ξ .*
- (b) *Is the system critical at zero temperature for antiferromagnetic couplings $J < 0$?*

- (c) Construct a renormalization group (RG) treatment by eliminating every other spin. Write down the recursion relations for the coupling J , and the additive constant g .
- (d) Discuss the fixed points, and their stability.
- (e) Write the expression for $\ln Z$ in terms of the additive constants of successive rescalings.
- (f) Show that the recursion relations are simplified when written in terms of $t(J) \equiv e^{-1/J}$.
- (g) Use the result in (f) to express the series in (e) in terms of t . Show that the answer can be reduced to that obtained in part (a), upon using the result

$$\sum_{n=0}^{\infty} \frac{1}{2^{n+1}} \ln \frac{1+t^{2^n}}{1-t^{2^n}} = -\ln(1-t).$$

- (h) Repeat the RG calculation of part (c), when a small symmetry breaking term $h \sum_i \delta_{s_i,1}$ is added to $-\beta\mathcal{H}$. You will find that an additional coupling term $K \sum_i \delta_{s_i,1} \delta_{s_{i+1},1}$ is generated under RG. Calculate the recursion relations in the three parameter space (J, K, h) .
- (i) Find the magnetic eigenvalues at the zero temperature fixed point where $J \rightarrow \infty$, and obtain the form of the correlation length close to zero temperature.

Solution:

(a)

Problem 6.9

Cluster RG: Consider Ising spins on a hexagonal lattice with nearest-neighbor interactions J .

- (a) Group the sites into clusters of four in preparation for a position space renormalization group with $b = 2$.
- (b) How can the majority rule be modified to define the renormalized spin of each cluster?
- (c) For a scheme in which the central site is chosen as the tie-breaker, make a table of all possible configurations of site-spins for a given value of the cluster-spin.
- (d) Focus on a pair of neighboring clusters. Indicate the contributions of intraccluster and intercluster bonds to the total energy.
- (e) Show that in zero magnetic field, the Boltzmann weights of parallel and anti-parallel clusters are given by

$$\begin{aligned} R_{++} &= x^8 + 2x^6 + 7x^4 + 14x^2 + 17 + 14x^{-2} + 7x^{-4} + 2x^{-6}, \\ R_{+-} &= 9x^4 + 16x^2 + 13 + 16x^{-2} + 9x^{-4} + x^{-8}, \end{aligned}$$

where $x = e^J$.

- (f) Find the expression for the resulting recursion relation $J'(J)$.
- (g) Estimate the critical ferromagnetic coupling J_c , and the exponent ν obtained from this RG scheme, and compare with the exact values.

- (h) What are the values of the magnetic and thermal exponents y_h , y_t at the zero temperature ferromagnetic fixed point?
- (i) Is the above scheme also applicable for antiferromagnetic interactions? What symmetry of the original problem is not respected by this transformation?

Solution:

(a)

Problem 6.10

Transition probability matrix: Consider a system of two Ising spins with a coupling K , which can thus be in one of four states.

- (a) Explicitly write the 4×4 transition matrix corresponding to single spin flips for a Metropolis algorithm. Verify that the equilibrium weights are indeed a left eigenvector of this matrix.
- (b) Repeat the above exercise if both single spin and double spin flips are allowed. The two types of moves are chosen randomly with probabilities p and $q = 1 - p$.

Solution:

(a)

Chapter 7

Series expansions

Problem 7.1

Continuous spins: In the standard $n\sigma$ model, n -component unit vectors are placed on the sites of a lattice. The nearest-neighbor spins are then connected by a bond $J\vec{s}_i \cdot \vec{s}_j$. In fact, if we are only interested in universal properties, any generalized interaction $f(\vec{s}_i \cdot \vec{s}_j)$ leads to the same critical behavior. By analogy with the Ising model, a suitable choice is

$$\exp(f(\vec{s}_i \cdot \vec{s}_j)) = 1 + nt \vec{s}_i \cdot \vec{s}_j,$$

resulting in the so-called loop model.

- (a) Construct a high-temperature expansion of the loop model (for the partition function Z) in the parameter t , on a two-dimensional hexagonal (honeycomb) lattice.
- (b) Show that the limit $n \rightarrow 0$ describes the configurations of a single self-avoiding polymer on the lattice.

Solution:

(a)

Problem 7.2

Potts model I: Consider Potts spins $s_i = 1, 2, \dots, q$, interacting via the Hamiltonian $-\beta\mathcal{H} = K \sum_{\langle ij \rangle} \delta_{s_i s_j}$.

- (a) To treat this problem graphically at high temperatures, the Boltzmann weight for each bond is written as

$$\exp(K\delta_{s_i s_j}) = C(K) [1 + T(K) g(s_i, s_j)],$$

with $g(s, s') = q\delta_{ss'} - 1$. Find $C(K)$ and $T(K)$.

- (b) Show that

$$\sum_{s=1}^q g(s, s') = 0, \quad \sum_{s=1}^q g(s_1, s)g(s, s_2) = q g(s_1, s_2), \quad \text{and} \quad \sum_{s, s'} g(s, s')g(s', s) = q^2(q-1).$$

- (c) Use the above results to calculate the free energy, and the correlation function $\langle g(s_m, s_n) \rangle$ for a one-dimensional chain.

- (d) Calculate the partition function on the square lattice to order of T^4 . Also calculate the first term in the low-temperature expansion of this problem.
- (e) By comparing the first terms in low- and high-temperature series, find a duality rule for Potts models. Assuming a single transition temperature, find the value of $K_c(q)$.
- (f) How do the higher-order terms in the high-temperature series for the Potts model differ from those of the Ising model? What is the fundamental difference that sets apart the graphs for $q = 2$? (This is ultimately the reason why only the Ising model is solvable.)

Solution:

(a)

Problem 7.3

Potts model II: An alternative expansion is obtained by starting with

$$\exp(K\delta_{s_i s_j}) = 1 + v(K)\delta_{s_i s_j},$$

where $v(K) = e^K - 1$. In this case, the sum over spins does not remove any graphs, and all choices of distributing bonds at random on the lattice are acceptable.

- (a) Including a magnetic field $h\sum_i \delta_{s_i,1}$, show that the partition function takes the form

$$Z(q, K, h) = \sum_{\text{all graphs}} \prod_{\text{clusters } c \text{ in graph}} \left[v_b^{n_b^c} \times (q - 1 + e^{hn_s^c}) \right],$$

where n_b^c and n_s^c are the numbers of bonds and sites in cluster c . This is known as the random cluster expansion.

- (b) Show that the limit $q \rightarrow 1$ describes a percolation problem, in which bonds are randomly distributed on the lattice with probability $p = v/(v+1)$. What is the percolation threshold on the square lattice?
- (c) Show that in the limit $q \rightarrow 0$, only a single connected cluster contributes to leading order. The enumeration of all such clusters is known as listing branched lattice animals.

Solution:

(a)

Problem 7.4

Ising model in a field: Consider the partition function for the Ising model ($\sigma_i = \pm 1$) on a square lattice, in a magnetic field h ; i.e.

$$Z = \sum_{\{\sigma_i\}} \exp \left(K \sum_{\langle ij \rangle} \sigma_i \sigma_j + h \sum_i \sigma_i \right).$$

- (a) Find the general behavior of the terms in a low-temperature expansion for Z .
- (b) Think of a model whose high-temperature series reproduces the generic behavior found in (a); and hence obtain the Hamiltonian, and interactions of the dual model.

Solution:

(a)

Problem 7.5

Potts duality: Consider Potts spins $s_i = 1, 2, \dots, q$, placed on the sites of a square lattice of N sites, interacting with their nearest-neighbors through a Hamiltonian $-\beta\mathcal{H} = K \sum_{\langle ij \rangle} \delta_{s_i s_j}$.

- (a) By comparing the first terms of high and low temperature series, or by any other method, show that the partition function has the property

$$Z(K) = qe^{2NK} \mathcal{F}(e^{-K}) = q^{-N} (e^K + q - 1)^{2N} \mathcal{F}\left(\frac{e^K - 1}{e^K + q - 1}\right),$$

for some function $\mathcal{F}$, and hence locate the critical point $K_c(q)$.

- (b) Starting from the duality expression for $Z(K)$, derive a similar relation for the internal energy $U(K) = \langle \mathcal{H} \rangle = -\partial \ln Z / \partial (\ln K)$. Use this to calculate the exact value of U at the critical point.

Solution:

(a)

Problem 7.6

Anisotropic random walks: Consider the ensemble of all random walks on a square lattice starting at the origin $(0,0)$. Each walk has a weight of $t_x^{\ell_x} \times t_y^{\ell_y}$, where ℓ_x and ℓ_y are the number of steps taken along the x and y directions, respectively.

- (a) Calculate the total weight $W(x, y)$ of all walks terminating at (x, y) . Show that W is well defined only for $t = (t_x + t_y)/2 < t_c = 1/4$.
- (b) What is the shape of a curve $W(x, y) = \text{constant}$, for large x and y , and close to the transition?
- (c) How does the average number of steps $\langle \ell \rangle = \langle \ell_x + \ell_y \rangle$ diverge as t approaches t_c ?

Solution:

(a)

Problem 7.7

Anisotropic Ising model: Consider the anisotropic Ising model on a square lattice with a Hamiltonian

$$-\beta\mathcal{H} = \sum_{x,y} [K_x \sigma_{x,y} \sigma_{x+1,y} + K_y \sigma_{x,y} \sigma_{x,y+1}],$$

i.e. with bonds of different strengths along the x and y directions.

- (a) *By following the method presented in the text, calculate the free energy for this model. You do not have to write down every step of the derivation. Just sketch the steps that need to be modified due to anisotropy; and calculate the final answer for $\ln Z/N$.*
- (b) *Find the critical boundary in the (K_x, K_y) plane from the singularity of the free energy. Show that it coincides with the condition $K_x = \tilde{K}_y$, where $\tilde{K}$ indicates the standard dual interaction to K .*
- (c) *Find the singular part of $\ln Z/N$, and comment on how anisotropy affects critical behavior in the exponent and amplitude ratios.*

Solution:

(a)

Problem 7.8

Müller–Hartmann and Zittartz estimate of the interfacial energy of the $d = 2$ Ising model on a square lattice:

- (a) *Consider an interface on the square lattice with periodic boundary conditions in one direction. Ignoring islands and overhangs, the configurations can be labelled by heights h_n for $1 \leq n \leq L$. Show that for an anisotropic Ising model of interactions K_x, K_y , the energy of an interface along the x -direction is*

$$-\beta\mathcal{H} = -2K_y L - 2K_x \sum_n |h_{n+1} - h_n|.$$

- (b) *Write down a column-to-column transfer matrix $\langle h|T|h' \rangle$, and diagonalize it.*
- (c) *Obtain the interface free energy using the result in (b), or by any other method.*
- (d) *Find the condition between K_x and K_y for which the interfacial free energy vanishes. Does this correspond to the critical boundary of the original 2d Ising model as computed in the previous problem?*

Solution:

(a)

Problem 7.9

Anisotropic Landau theory: Consider an n -component magnetization field $\vec{m}(\vec{x})$ in d dimensions.

- (a) *Using the previous problems on anisotropy as a guide, generalize the standard Landau–Ginzburg Hamiltonian to include the effects of spatial anisotropy.*
- (b) *Are such anisotropies “relevant”?*
- (c) *In La_2CuO_4 , the Cu atoms are arranged on the sites of a square lattice in planes, and the planes are then stacked together. Each Cu atom carries a “spin”, which we assume to be classical, and can point along any direction in space. There is a very strong anti-ferromagnetic interaction in each plane. There is also a very weak interplane interaction that prefers to align successive layers. Sketch the low-temperature magnetic phase, and indicate to what universality class the order–disorder transition belongs.*

Solution:

(a)

Problem 7.10

Energy by duality: Consider the Ising model ($\sigma_i = \pm 1$) on a square lattice with $-\beta\mathcal{H} = K \sum_{\langle ij \rangle} \sigma_i \sigma_j$.

- (a) *Starting from the duality expression for the free energy, derive a similar relation for the internal energy $U(K) = \langle \mathcal{H} \rangle = -\partial \ln Z / \partial (\ln K)$.*
- (b) *Using (a), calculate the exact value of U at the critical point K_c .*

Solution:

(a)

Problem 7.11

Clock model duality: Consider spins $s_i = 1, 2, \dots, q$ placed on the sites of a square lattice, interacting via the clock model Hamiltonian $\mathcal{H}_C = -\sum_{\langle ij \rangle} J(s_i - s_j \bmod q)$.

- (a) *Change from the N site variables to the $2N$ bond variables $b_{ij} = s_i - s_j$. Show that the difference in the number of variables can be accounted for by the constraint that around each plaquette (elementary square) the sum of the four bond variables must be zero modulus q .*

(b) The constraints can be implemented by adding “delta-functions”

$$\delta\left(\sum_{\text{plaq}} S_p \bmod q\right) = \frac{1}{q} \sum_{n_p=1}^q \exp\left(\frac{2\pi i n_p S_p}{q}\right)$$

for each plaquette. Show that after summing over the bond variables, the partition function can be written in terms of the dual variables, as

$$Z = q^{-N} \sum_{\{n_p\}} \prod_{\langle pp' \rangle} \tilde{\lambda}(n_p - n_{p'}) \equiv \sum_{\{n_p\}} \exp \left[\sum_{\langle pp' \rangle} \tilde{J}(n_p - n_{p'}) \right],$$

where $\tilde{\lambda}(k)$ is the discrete Fourier transform of $e^{J(n)}$.

(c) Calculate the dual interaction parameter of a Potts model, and hence locate the critical point $J_c(q)$.

(d) Construct the dual of the anisotropic Potts model, with

$$-\beta\mathcal{H} = \sum_{x,y} [J_x \delta_{s_{x,y}, s_{x+1,y}} + J_y \delta_{s_{x,y}, s_{x,y+1}}],$$

i.e. with bonds of different strengths along the x and y directions. Find the line of self-dual interactions in the plane (J_x, J_y) .

Solution:

(a)

Problem 7.12

Triangular/hexagonal lattices: For any planar network of bonds, one can define a geometrical dual by connecting the centers of neighboring plaquettes. Each bond of the dual lattice crosses a bond of the original lattice, allowing for a local mapping. Clearly, the dual of a triangular lattice is a hexagonal (or honeycomb) lattice, and vice versa.

(a) Starting from a Potts model with nearest-neighbor interaction J on the triangular lattice, find the interaction parameter $J_h(J)$ of the dual hexagonal lattice.

(b) Note that the hexagonal lattice is bipartite, i.e. can be separated into two sublattices. In the partition function, do a partial sum over all spins in one sublattice. Show that the remaining spins form a triangular lattice with nearest-neighbor interactions $\tilde{J}(J)$, and a three-spin interaction $\tilde{J}_3(J)$. (This is called the star-triangle transformation.) Why is $\tilde{J}_3$ absent in the Ising model?

(c) Clearly, the model is not self-dual due to the additional interaction. Nonetheless, obtain the critical value such that $\tilde{J}(J_c) = J_c$.

(d) Check that $\tilde{J}_3(J_c) = 0$, i.e. while the model in general is not self-dual, it is self-dual right at criticality, leading to the exact value of $J_c(q)$!

Solution:

(a)

Problem 7.13

Cubic lattice: The geometric concept of duality can be extended to general dimensions d . However, the dual of a geometric element of dimension D is an entity of dimension $d - D$. For example, the dual of a bond ($D = 1$) in $d = 3$ is a plaquette ($D = 2$), as demonstrated in this problem.

- (a) *Consider a clock model on a cubic lattice of N points. Change to the $3N$ bond variables $b_{ij} = s_i - s_j$. Show that there are now $2N$ constraints associated with the plaquettes of this lattice.*
- (b) *Implement the constraints through discrete delta-functions by associating an auxiliary variable n_p with each plaquette. It is useful to imagine n_p as defined on a bond of the dual lattice, perpendicular to the plaquette p .*
- (c) *By summing over the bond variables in Z , obtain the dual Hamiltonian*

$$\tilde{\mathcal{H}} = \sum_p \tilde{J}(n_{12}^p - n_{23}^p + n_{34}^p - n_{41}^p),$$

where the sum is over all plaquettes p of the dual lattice, with $\{n_{ij}^p\}$ indicating the four bonds around plaquette p .

- (d) *Note that $\tilde{\mathcal{H}}$ is left invariant if all the six bonds going out of any site are simultaneously increased by the same integer. Thus unlike the original model which only had a global translation symmetry, the dual model has a local, i.e. gauge symmetry.*
- (e) *Consider a Potts gauge theory defined on the plaquettes of a four-dimensional hypercubic lattice. Find its critical coupling $J_c(q)$.*

Solution:

(a)

Percolation

Problem 7.14

Cayley trees: Consider a hierarchical lattice in which each site at one level is connected to z sites at the level below. Thus the n th level of the tree has z^n sites.

- (a) *For $z = 2$, obtain a recursion relation for the probability $P_n(p)$ that the top site of a tree of n levels is connected to some site at the bottom level.*
- (b) *Find the limiting behavior of $P_\infty(p)$ for infinitely many levels, and give the exponent β for this tree.*

- (c) In a mean-field approximation, $P_\infty(p)$ for a lattice of coordination number z is calculated self-consistently, in a manner similar to the above. Give the mean-field estimates for p_c and the exponent β .
- (d) The one-dimensional chain corresponds to $z = 1$. Find the probability that endpoints of an open chain of $N + 1$ sites are connected, and hence deduce the correlation length ξ .

Solution:

(a)

Problem 7.15

Duality in percolation: Duality has a very natural interpretation in percolation: if a bond is occupied, its dual is empty, and vice versa. Thus the occupation probability for dual bonds is $\tilde{p} = 1 - p \equiv q$.

- (a) The dual of a chain in which N bonds are connected in series has N bonds connected in parallel. What is the corresponding (non-)percolation probability?
- (b) The bond percolation problem on a square lattice is self-dual. What is its threshold p_c ?
- (c) Bond percolation in three dimensions is dual to plaquette percolation. This is why in $d = 3$ it is possible to percolate while maintaining solid integrity.
- (d) The triangular and hexagonal lattices are dual to each other. Combined with the star-triangle transformation that also maps the two lattices, this leads to exact values of p_c for these lattices. However, this calculation is not trivial, and it is best to follow the steps for calculations of the critical couplings of q -state Potts models in an earlier problem.

Solution:

(a)

Problem 7.16

Position-space renormalization group (PSRG): In the Migdal-Kadanoff approximation, a decimation is performed after bonds are moved to connect the remaining sites in a one-dimensional geometry. For a PSRG rescaling factor b on a d -dimensional hypercubic lattice, the retained sites are connected by b^{d-1} strands in parallel, each strand having b bonds in series.

- (a) Apply the above scheme to $b = d = 2$, and compare the resulting estimates of p_c and $y_t = 1/\nu$ to the exact values for the square lattice ($p_c = 1/2$ and $y_t = 3/4$).
- (b) Find the limiting values of p_c and y_t for large d . (It can be shown that $\nu = 1/2$ exactly, above an upper critical dimension $d_u = 6$.)
- (c) In an alternative PSRG scheme, first b^{d-1} bonds are connected in parallel, the resulting groups are then joined in series. Repeat the above calculations with this scheme.

Solution:

(a)

Chapter 8

Beyond spin waves

Problem 8.1

Anisotropic nonlinear σ model: Consider unit n -component spins $\vec{s}(\vec{x}) = (s_1, \dots, s_n)$ with $\sum_{\alpha} s_{\alpha}^2 = 1$, subject to a Hamiltonian

$$\mathcal{H} = \int d^d x \left[\frac{1}{2T} (\nabla \vec{s})^2 + g s_1^2 \right].$$

For $g = 0$, renormalization group equations are obtained through rescaling distances by a factor $b = e^{\ell}$, and spins by a factor $\zeta = b^{y_s}$ with $y_s = -(n-1)T/(4\pi)$, and lead to the flow equation

$$\frac{dT}{d\ell} = -\epsilon T + \frac{n-2}{2\pi} T^2 + \mathcal{O}(T^3),$$

where $\epsilon = d - 2$.

- (a) *Find the fixed point, and the thermal eigenvalue y_T .*
- (b) *Write the renormalization group equation for g in the vicinity of the above fixed point, and obtain the corresponding eigenvalue y_g .*
- (c) *Sketch the phase diagram as a function of T and g , indicating the phases, and paying careful attention to the shape of the phase boundary as $g \rightarrow 0$.*

Solution:

(a)

Problem 8.2

Matrix models: In some situations, the order parameter is a matrix rather than a vector. For example, in triangular (Heisenberg) antiferromagnets each triplet of spins aligns at 120° , locally defining a plane. The variations of this plane across the system are described by a 3×3 rotation matrix. We can construct a nonlinear σ model to describe a generalization of this problem as follows. Consider the Hamiltonian

$$\mathcal{H} = \frac{K}{4} \int d^d x \operatorname{tr} [\nabla M(\vec{x}) \cdot \nabla M^T(\vec{x})],$$

where M is a real, $N \times N$ orthogonal matrix, and “tr” denotes the trace operation. The condition of orthogonality is that $MM^T = M^T M = I$, where I is the $N \times N$ identity matrix, and M^T is the transposed matrix, $M_{ij}^T = M_{ji}$.

- (a) Rewrite the Hamiltonian and the orthogonality constraint in terms of the matrix elements M_{ij} ($i, j = 1, \dots, N$). Describe the ground state of the system.
- (b) Define the symmetric and anti-symmetric matrices

$$\Sigma = \frac{1}{2}(M + M^T) = \Sigma^T, \quad \Pi = \frac{1}{2}(M - M^T) = -\Pi^T.$$

Express $\mathcal{H}$ and the orthogonality constraint in terms of the matrices Σ and Π .

- (c) Consider small fluctuations about the ordered state $M(\vec{x}) = I$. Show that Σ can be expanded in powers of Π as

$$\Sigma = I - \frac{1}{2}\Pi\Pi^T + \dots$$

Use the orthogonality constraint to integrate out Σ , and obtain an expression for $\mathcal{H}$ to fourth order in Π . Note that there are two distinct types of fourth-order terms. Do not include terms generated by the argument of the delta function.

- (d) For an N -vector order parameter there are $N - 1$ Goldstone modes. Show that an orthogonal $N \times N$ order parameter leads to $N(N - 1)/2$ such modes.
- (e) Consider the quadratic piece of $\mathcal{H}$. Show that the two-point correlation function in Fourier space is

$$\langle \Pi_{ij}(\vec{q}) \Pi_{kl}(\vec{q}') \rangle = \frac{(2\pi)^d \delta^d(\vec{q} + \vec{q}')}{K q^2} (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}).$$

- (f) Calculate the coarse-grained expectation value for $\text{tr}\langle \Sigma^> \rangle_0$ at low temperatures after removing the shell of Fourier modes $M^>(\vec{q})$ with $q \in (\Lambda/b, \Lambda)$. Identify the scaling factor $M'(\vec{x}') = M^<(\vec{x})/\zeta$ that restores $\text{tr}M' = \text{tr}\Sigma' = N$.
- (g) Use perturbation theory to calculate the coarse-grained coupling constant $\tilde{K}$. Evaluate only the two diagrams that directly renormalize the Π_{ij}^2 term in $\mathcal{H}$, and show that

$$\tilde{K} = K + \frac{N}{2} \int_{\Lambda/b}^{\Lambda} \frac{d^d q}{(2\pi)^d} \frac{1}{q^2}.$$

- (h) Using the result from part (f), show that after matrix rescaling, the RG equation for K' is given by:

$$K' = b^{d-2} \left(K - \frac{N-2}{2} \int_{\Lambda/b}^{\Lambda} \frac{d^d q}{(2\pi)^d} \frac{1}{q^2} \right).$$

- (i) Obtain the differential RG equation for $T = 1/K$, by considering $b = 1 + d\ell$. Sketch the flows for $d < 2$ and $d = 2$. For $d = 2 + \epsilon$, compute T_c and the critical exponent ν .
- (j) Consider a small symmetry breaking term $-h \int d^d x \text{tr} M$, added to the Hamiltonian. Find the renormalization of h , and identify the corresponding exponent y_h .

Solution:

(a)

Problem 8.3

The roughening transition: In an earlier problem we examined a continuum interface model which in $d = 3$ is described by the Hamiltonian

$$\mathcal{H}_0 = \frac{K}{2} \int d^2x (\nabla h)^2,$$

where $h(\vec{x})$ is the interface height at location $\vec{x}$. For a crystalline facet, the allowed values of h are multiples of the lattice spacing. In the continuum, this tendency for integer h can be mimicked by adding a term

$$-\mathcal{U} = y_0 \int d^2x \cos(2\pi h)$$

to the Hamiltonian. Treat $-\mathcal{U}$ as a perturbation, and proceed to construct a renormalization group as follows:

(a) *Show that*

$$\langle \exp(i\lambda h(\vec{x})) \rangle_0 = \exp\left(\frac{\lambda^2}{K} \int^< \frac{d^2q}{(2\pi)^2} C(\vec{q}; \vec{x} - \vec{x}')\right)$$

for $\lambda \neq 0$, and zero otherwise, where $C(\vec{x}) = \ln|\vec{x}|/(2\pi)$ is the Coulomb interaction in two dimensions.

(b) *Prove that*

$$\langle (h(\vec{x}) - h(\vec{y}))^2 \rangle = - \left. \frac{d^2}{dk^2} G_k(\vec{x} - \vec{y}) \right|_{k=0},$$

where $G_k(\vec{x} - \vec{y}) = \langle \exp(ik(h(\vec{x}) - h(\vec{y}))) \rangle$.

(c) *Use the results in (a) to calculate $G_k(\vec{x} - \vec{y})$ in perturbation theory to order of y_0^2 . (Set $\cos(2\pi h) = (e^{2\pi i h} + e^{-2\pi i h})/2$. The first-order terms vanish according to the result in (a), while the second-order contribution is identical in structure to that of the Coulomb gas described in this chapter.)*

(d) *Write the perturbation result in terms of an effective interaction K , and show that perturbation theory fails for K larger than a critical K_c .*

(e) *Recast the perturbation result in part (d) into renormalization group equations for K and y_0 , by changing the “lattice spacing” from a to ae^ℓ .*

(f) *Using the recursion relations, discuss the phase diagram and phases of this model.*

(g) *For large separations $|\vec{x} - \vec{y}|$, find the magnitude of the discontinuous jump in $\langle (h(\vec{x}) - h(\vec{y}))^2 \rangle$ at the transition.*

Solution:

(a)

Problem 8.4

Roughening and duality: Consider a discretized version of the Hamiltonian in the previous problem, in which for each site i of a square lattice there is an integer-valued height h_i . The Hamiltonian is

$$\mathcal{H} = \frac{K}{2} \sum_{\langle ij \rangle} |h_i - h_j|^2,$$

where the $*$ power means that there is no energy cost for $\Delta h = 0$; an energy cost of $K/2$ for $\Delta h = \pm 1$; and $\Delta h = \pm 2$ or higher are not allowed for neighboring sites. (This is known as the restricted solid on solid (RSOS) model.)

- (a) Construct the dual model either diagrammatically, or by following these steps:
- (i) Change from the N site variables h_i , to the $2N$ bond variables $n_{ij} = h_i - h_j$. Show that the sum of n_{ij} around any plaquette is constrained to be zero.
 - (ii) Impose the constraints by using the identity $\int_0^{2\pi} \frac{d\theta}{2\pi} e^{in\theta/2} = \delta_{n,0}$ for integer n .
 - (iii) After imposing the constraints, sum freely over the bond variables n_{ij} to obtain a dual interaction $\tilde{v}(\theta_i - \theta_j)$ between dual variables θ_i on neighboring plaquettes.
- (b) Show that for large K , the dual problem is just the XY model. Is this conclusion consistent with the renormalization group results of the previous problem?
- (c) Does the one-dimensional version of this Hamiltonian, i.e. a 2d interface with $-\mathcal{H} = -(K/2) \sum_i |h_i - h_{i+1}|^*$, have a roughening transition?

Solution:

(a)

Problem 8.5

Nonlinear σ model with long-range interactions: Consider unit n -component spins $\vec{s}(\vec{x}) = (s_1, s_2, \dots, s_n)$ with $|\vec{s}(\vec{x})|^2 = 1$, interacting via a Hamiltonian

$$\mathcal{H} = \int d^d x d^d y K(\vec{x} - \vec{y}) \vec{s}(\vec{x}) \cdot \vec{s}(\vec{y}).$$

- (a) The long-range interaction $K(\vec{x})$ is the Fourier transform of $K(\vec{q}) \propto q^{\sigma/2}$ with $\sigma < 2$. What kind of asymptotic decay of interactions at long distances is consistent with such behavior?
- (b) Close to zero temperature we can set $\vec{s}(\vec{x}) = (\vec{\pi}(\vec{x}), s_n(\vec{x}))$ where $\vec{\pi}(\vec{x})$ is an $(n-1)$ -component vector representing small fluctuations around the ground state. Find the effective Hamiltonian for $\vec{\pi}(\vec{x})$ after integrating out $s_n(\vec{x})$.
- (c) Fourier transform the quadratic part of the above Hamiltonian focusing only on terms proportional to K , and hence calculate the expectation value $\langle \pi_{i,\vec{q}} \pi_{j,\vec{q}} \rangle_0$.
- (d) Calculate the coarse-grained expectation value for $\langle s_n^{\geq} \rangle_0$ to order of π^2 after removing the modes $\vec{\pi}^{\geq}(\vec{q})$ with $q \in (\Lambda/b, \Lambda)$. Identify the scaling factor $\vec{s}'(\vec{x}') = \vec{s}^{\leq}(\vec{x})/\zeta$ that restores $|\vec{s}'|$ to unit length.
- (e) A simplifying feature of long-range interactions is that the coarse-grained coupling constant is not modified by the perturbation, i.e. $\tilde{K} = K$ to all orders in a perturbative calculation. Use this information, along with simple dimensional analysis, to express the renormalized interaction $K'(b)$ in terms of K , b , and σ .
- (f) Obtain the differential RG equation for $T = 1/K$ by considering $b = 1 + d\ell$.
- (g) For $d = \sigma + \epsilon$, compute T_c and the critical exponent ν to lowest order in ϵ .

- (h) Add a small symmetry breaking term $-\vec{h} \cdot \int d^d x \vec{s}(\vec{x})$ to the Hamiltonian. Find the renormalization of h and identify the corresponding exponent y_h .

Solution:

(a)

Problem 8.6

The XY model in $2 + \epsilon$ dimensions: The recursion relations of the XY model in two dimensions can be generalized to $d = 2 + \epsilon$ dimensions, and take the form

$$\begin{cases} \frac{dT}{d\ell} = -\epsilon T + 4\pi^3 y^2, \\ \frac{dy}{d\ell} = \left(2 - \frac{\pi}{T}\right) y. \end{cases}$$

- (a) Calculate the position of the fixed point for the finite-temperature phase transition.
(b) Obtain the eigenvalues at this fixed point to lowest contributing order in ϵ .
(c) Estimate the exponents ν and η for the superfluid transition in $d = 3$ from these results. [Be careful in keeping track of only the lowest nontrivial power of ϵ in your expressions.]

Solution:

(a)

Problem 8.7

Symmetry breaking fields: Let us investigate adding a term

$$-\mathcal{H}_p = h_p \int d^2 x \cos(p\theta(\vec{x}))$$

to the XY model. There are a number of possible causes for such a symmetry breaking field: $p = 1$ is the usual “magnetic field”, $p = 2, 3, 4$, and 6 could be due to couplings to an underlying lattice of rectangular, hexagonal, square, or triangular symmetry respectively.

- (a) Assume that we are in the low-temperature phase so that vortices are absent, i.e. the vortex fugacity y is zero (in the RG sense). In this case, we can ignore the angular nature of θ and replace it with a scalar field ϕ , leading to the partition function

$$Z = \int \mathcal{D}\phi(\vec{x}) \exp\left(-\int d^2 x \left[\frac{K}{2}(\nabla\phi)^2 + h_p \cos(p\phi)\right]\right).$$

This is known as the sine-Gordon model. Use similar methods to obtain the recursion relations for h_p and K .

- (b) Show that once vortices are included, the recursion relations are

$$\frac{dh_p}{d\ell} = \left(2 - \frac{p^2}{4\pi K}\right) h_p, \quad \frac{dK^{-1}}{d\ell} = -\frac{\pi p^2 h_p^2}{2} K^{-2} + 4\pi^3 y^2, \quad \frac{dy}{d\ell} = \left(2 - \frac{\pi}{K}\right) y.$$

- (c) Show that the above RG equations are only valid for $8\pi/p^2 < K^{-1} < \pi/2$, and thus only apply for $p > 4$. Sketch possible phase diagrams for $p > 4$ and $p < 4$.

Solution:

(a)

Problem 8.8

Inverse-square interactions: Consider a scalar field $s(\vec{x})$ in one dimension, subject to an energy

$$-\mathcal{H}_s = \frac{K}{2} \int dx dy \frac{s(x)s(y)}{(x-y)^2} + \int dx \Phi(s),$$

where the local energy $\Phi(s)$ strongly favors $s(x) = \pm 1$ (e.g. $\Phi(s) = g(s^2 - 1)^2$, with $g \gg 1$).

- (a) With $K > 0$, the ground state is ferromagnetic. Estimate the energy cost of a single domain wall in a chain of length L . You may assume that the transition from $s = +1$ to $s = -1$ occurs over a short distance cutoff a .
- (b) From the probability of the formation of a single kink, obtain a lower bound for the critical coupling K_c , separating ordered and disordered phases.
- (c) Show that the energy of a dilute set of domain walls located at positions x_i is given by

$$-\mathcal{H}_Q = 4K \sum_{i < j} q_i q_j \ln \frac{|x_i - x_j|}{a} + \ln y_0 \sum_i |q_i|,$$

where $q_i = \pm 1$ depending on whether $s(x)$ increases or decreases at the domain wall. (y_0 is the fugacity of a single domain wall.)

- (d) The logarithmic interaction between two opposite domain walls at a large distance L is reduced due to screening by other domain walls in between. This interaction can be calculated perturbatively in y_0 , and to lowest order is described by an effective coupling

$$K \rightarrow K_{\text{eff}} = K - 2Ky_0^2 \int_a^\infty dr r \left(\frac{a}{r}\right)^{4K} + \mathcal{O}(y_0^4). \quad (1)$$

By changing the cutoff from a to $ba = (1 + d\ell)a$, construct differential recursion relations for the parameters K and y_0 .

- (e) Sketch the renormalization group flows as a function of $T = K^{-1}$ and y_0 , and discuss the phases of the model.
- (f) Derive the effective interaction given above as Eq. (1).

Solution:

(a)

Problem 8.9

Melting: The elastic energy cost of a deformation $u_i(\vec{x})$ of an isotropic lattice is

$$-\mathcal{H} = \frac{1}{2} \int d^d x (2\mu u_{ij}u_{ij} + \lambda u_{ii}u_{jj}),$$

where $u_{ij}(\vec{x}) = (\partial_i u_j + \partial_j u_i)/2$ is the strain tensor.

- (a) *Express the energy in terms of the Fourier transforms $u_i(\vec{q})$, and find the normal modes of vibrations.*
- (b) *Calculate the expectation value $\langle u_i(\vec{q}) u_j(\vec{q}') \rangle$.*
- (c) *Assuming a short-distance cutoff of the order of the lattice spacing a , calculate $U^2(\vec{x}) \equiv \langle (\vec{u}(\vec{x}) - \vec{u}(\vec{0}))^2 \rangle$.*
- (d) *The (heuristic) Lindemann criterion states that the lattice melts when deformations grow to a fraction of the lattice spacing, i.e. for $\lim_{x \rightarrow \infty} U(x) = c_L a$. Assuming that $\mu = \hat{\mu}/k_B T$ and $\lambda = \hat{\lambda}/k_B T$, use the above criterion to calculate the melting temperature T_m . Comment on the behavior of T_m as a function of dimension d .*

Solution:

(a)
